R-UNIVERSALITY OF THE ABSOLUTELY SIMPLE ALGEBRAS OF TYPE E7 DETERMINED BY A SINGLE TITS ALGEBRA

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Allowed standard prerequisites (editorial).

- The fact that a simply connected group of type E7 has centre of order two, so that only one Tits algebra occurs. Used at the start of the counted proof (source line 834). Reduces R-universality to the behaviour of that single algebra.
- The classification of type E7 forms by their Tits index, with the tabulated strong innerness and anisotropicity of each index. Used inside the counted proof at source lines 836-841, attributed by the author to section 6.5.5 of Tits-RepLinIrred and to propositions 4 and 5 of Tits-StronglyInner. Supplies which indices have trivial Tits algebra and which are anisotropic, which is exactly what the two alternatives record.

Proposition 0.1. *Let $\mathfrak{g}$ be an absolutely simple Lie algebra over $\mathbb{Q}$ of type E_7. Then $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if either*

(1) *$\mathfrak{g}$ is of type $E_{7,3}^{28}$ or $E_{7,7}^{0}$ (over $\mathbb{Q}$), or*
(2) *$\mathfrak{g} \otimes_{\mathbb{Q}} \mathbb{R}$ is of type $E_{7,0}^{133}$, $E_{7,2}^{31}$, or $E_{7,4}^{9}$.*

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Proof. Since a simply connected group of type E_7 has a center of order 2, there is only one Tits algebra to consider. For $\mathfrak{g}$ to be $\mathbb{R}$-universal, this algebra needs to either be trivial over $\mathbb{Q}$ or nontrivial over $\mathbb{R}$. Tits [11, § 6.5.5] points out that it is trivial (over any field) for types $E_{7,3}^{28}$ and $E_{7,7}^{0}$, but nontrivial (over any field) for types $E_{7,2}^{31}$ and $E_{7,4}^{9}$. Since there are no strongly inner anisotropic groups of type E_7 over $\mathbb{Q}$ or $\mathbb{R}$ [12, Propositions 4 and 5], the Tits algebra is also nontrivial for the anisotropic Lie algebra $E_{7,0}^{133}$ (over the fields of interest to us). □

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